PLACE VALUE STUDIO & REGROUPING TRAYS

Dual Whole Number & Decimal Activity Trays with Base-Ten Regrouping Tokens

SERIES 1 • CARD 12 / 12 • DOMAIN: POSITIONAL BASE-10 & REGROUPING

CARD 12 / 12 • PLACE-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

POSITIONAL BASE-10 NOTATION & REGROUPING

A digit's value depends on its place column: $d_k \times 10^k$.

Addition: Trade 10 units for 1 Carry block (+1) one column left.

Subtraction: Borrow 1 from left column for a (10) Trade block.

Dual 5x2 trays connect whole numbers directly to decimals!

CALIBRATED 19.00 MM POSITIONAL GRID

- Digit Tile: 19.00 × 19.00 × 7.00 mm (1U) with 1.85mm detent
- Whole Tray: 10,000, 1000, 100, 10, 1 (Inner 5×2 Tray #1)
- Decimal Tray: 0.1, 0.01, 0.001, 10^{-4}, 10^{-5} (Inner 5×2 Tray #2)

PLACE VALUE REGROUPING: 347 + 285 = 632

THOU	HUND	TENS	ONES
	3	+1 4	7
	2	8	5

REGROUPING TOKENS

Trade 10 Ones for 1 Ten

7 + 5 = 12 → 2 ones, +1 carry

Dual 6x2 Trays:

Whole: Thousands to Ones

Decimal: Ones to Thousandths

2. KIT PARTS INVENTORY (56 PIECES)

COMPONENT	QTY	SPEC	COLOR
White Digits (0–9)	22x	4×0, 2×1–9	PLA White
Blue Digits (0–9)	22x	4×0, 2×1–9	Blue-Gray
Carry & Borrow Tokens	8x	4×, 4×	Glow Yellow
Whole Numbers Tray	1x	Inner 5×2	Blue-Gray
Decimal Numbers Tray	1x	Inner 5×2	Blue-Gray
3x1 Storage Bin	1x	49.0 mm	PLA White
[123] Docking Lid	1x	Docking Lid	PLA White

TOTAL: 44 Digit Tiles + 8 (+1/10) Tokens + 2 Inner 5x2 Trays + Box + Lid = 56 Parts

SYMMETRIC 2-COLOR OPERAND & REGROUPING PALETTE

- PLA White (Set A, 22x): 4x '0' + 2x each of '1'–'9' + Box + Lid.
- Slate Blue-Gray (Set B, 22x): 4x '0' + 2x each of '1'–'9'.
- Slate Blue-Gray (2x Inner Trays): 5×2 Whole & Decimal Trays.
- Glow Yellow / Red (8x): 4x (+1) Carry + 4x (10) Borrow tokens.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Stack 4 blocks in Cols 0–4 (0–9), stack 6 blocks in Col 5 (+1, 10, & extra 0s = 52 blocks total), nest the 2 inner 5x2 trays over Cols 0–4 (leaving Col 5 for easy finger grip)

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

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Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every real number has a unique base-10 positional representation.

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LESSON GUIDE: MULTI-DIGIT ADDITION & DECIMAL REGROUPING

Step-by-Step Hands-On Tutorial: 347 + 285 = 632 with Regrouping Tokens

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

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Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 2.NBT.A.1 (Place Value)
- CCSS 4.NBT.B.4 (Addition/Sub...)
- CCSS 5.NBT.A.3 (Decimals to ...)

Grades 1–5 Elementary Mathematics

GUIDING INQUIRY

What happens when the counters in any column reach ten?

1 STEP 1: POPULATE OPERANDS 347 AND 285

Place 347 in top row (White) and 285 in bottom row (Tan).
Align 3 hundreds, 4 tens, 7 ones over 2 hundreds, 8 tens, 5 ones.
Visual clarity: operand contrast eliminates confusion.

2 STEP 2: REGROUP ONES (7 + 5 = 12)

7 + 5 = 12. Trade 10 ones for a Red (+1) token.
Place the Red token on the Tens column digit block.
Keep 2 in the Ones column.

3–4 STEPS 3 & 4: REGROUP TENS & COMPLETE

3. REGROUP TENS: 4 + 8 + 1 = 13. Trade 10 tens for a Red (+1) in Hundreds.
4. READ COMPLETED SUM: Keep 3 in Tens; Hundreds: 3 + 2 + 1 = 6. Total = 632!

5 CORE PRINCIPLE: CARRYING IS PHYSICAL TRADING

In standard algorithms, carrying is a mysterious tiny 1 written above a column.
Here, carrying is a physical trade: 10 units become 1 ten block.
Students see that value is conserved across columns!

★ EXTENSION INVESTIGATION: DECIMAL CHALLENGE: 1.75 + 0.85

Set up Decimal tray with 1.75 and 0.85. Regroup hundredths and tenths. What is the sum?
2.60! (Scan QR for 3D lesson!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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